in the second variable with the same Lipschitz constant $\alpha < 1$. For $\boldsymbol{p} \in P$, let $\boldsymbol{x}^*(\boldsymbol{p})$ be the unique fixed point of the map $\boldsymbol{x} \to F(\boldsymbol{p}, \boldsymbol{x})$. Then the mapping $\boldsymbol{x}^*$ is continuous.

This theorem is Theorem 2.3 in Frigon, 2007, [27]. Corollary 4.2 in Feinstein, 2016, generalized the theorem to set valued operators, that is, these operators may have more than one fixed point [25] (see also [67]). All mappings $F(p, .)$ must have the same Lipschitz constant $\alpha < 1$.

A locally compact space is a space where every point has a compact neighborhood. $\mathbb{R}^n$ is locally compact as a consequence of the Heine-Borel theorem. Proposition 3.2 in Jachymski, 1996, is [62]:

Theorem A12 (Jachymski: locally compact complete metric space)**.** *Let (X, d) be a locally compact complete metric space, and let (P, d_P) be a metric space. Let $F : P \times X \to X$ be continuous in the first variable and contractive in the second variable with not necessarily the same Lipschitz constant. For $\boldsymbol{p} \in P$, let $\boldsymbol{x}^*(\boldsymbol{p})$ be the unique fixed point of the map $\boldsymbol{x} \to F(\boldsymbol{p}, \boldsymbol{x})$. Then the mapping $\boldsymbol{x}^*$ is continuous.*

This theorem is Theorem 2.5 in Frigon, 2007, [27] and Theorem 2 in Kwiecinski, 1992, [70]. The mappings $F(p, .)$ can have different Lipschitz constants.

A7.1.5 t-fold Composition of the Operator

We define the Bellman operator as

$$\begin{aligned} \mathrm{T}^\pi\,[q]\,(s,a) \;&=\; g(s,a) \;+\; \sum_{s'} p(s' \mid s,a) \sum_{a'} \pi(a' \mid s')\, q(s',a') \\ &=\; g(s,a) \;+\; \boldsymbol{q}^T \boldsymbol{p}(s,a)\;, \end{aligned} \tag{A292}$$

where $\boldsymbol{q}$ is the vector with value $q(s', a')$ at position (s', a') and $\boldsymbol{p}(s, a)$ is the vector with value $p(s' \mid s, a)\pi(a' \mid s')$ at position (s', a').

In vector notation we obtain the *Bellman equation* or *Poisson equation*. For the Poisson equation see Equation 33 to Equation 37 for the undiscounted case and Equation 34 and Equation 43 for the discounted case in Alexander Veretennikov, 2016, [137]. This form of the Poisson equation describes the Dirichlet boundary value problem. The *Bellman equation* or *Poisson equation* is

$$\mathrm{T}^\pi\,[\boldsymbol{q}] \;=\; \boldsymbol{g} \;+\; \boldsymbol{P}\,\boldsymbol{q}\;, \tag{A293}$$

where $\boldsymbol{P}$ is the row-stochastic matrix with $p(s' \mid s, a)\pi(a' \mid s')$ at position $((s, a), (s', a'))$.

The Poisson equation is

$$\boldsymbol{q}^\pi \;+\; \bar{g}\mathbf{1} \;=\; \boldsymbol{g} \;+\; \boldsymbol{P}\,\boldsymbol{q}\;, \tag{A294}$$

where $\mathbf{1}$ is the vector of ones and $\bar{\boldsymbol{g}}$ is the long term average reward or the expected value of the reward for the stationary distribution:

$$\bar{g} \;=\; \lim_{T\to\infty} \frac{1}{T+1} \sum_{t=0}^{T} g(s_t, a_t)\;. \tag{A295}$$

We assume $\bar{g} = 0$ since after some time the agent does no longer receive reward for MDPs with finite time horizon or MDPs with absorbing states that have zero reward.

Since $\boldsymbol{P}$ is a row-stochastic matrix, the Perron-Frobenius theorem says that (1) $\boldsymbol{P}$ has as largest eigenvalue 1 for which the eigenvector corresponds to the steady state and (2) the absolute value of each (complex) eigenvalue is smaller or equal 1. Only the eigenvector to the eigenvalue 1 has purely positive real components.

Equation 7 of Bertsekas and Tsitsiklis, 1991, [13] states

$$(\mathrm{T}^\pi)^t\,[\boldsymbol{q}] \;=\; \sum_{k=0}^{t-1} \boldsymbol{P}^k\,\boldsymbol{g} \;+\; \boldsymbol{P}^t\,\boldsymbol{q}\;. \tag{A296}$$

If $\boldsymbol{p}$ is the stationary distribution vector for $\boldsymbol{P}$, that is,

$$\lim_{k\to\infty} \boldsymbol{P}^k \;=\; \mathbf{1}\,\boldsymbol{p}^T \tag{A297}$$

$$\lim_{k\to\infty} \boldsymbol{p}_0^T \boldsymbol{P}^k \;=\; \boldsymbol{p}^T \tag{A298}$$

then

$$\lim_{k\to\infty} \frac{1}{k} \sum_{i=0}^{k-1} \boldsymbol{P}^i \;=\; \mathbf{1}\,\boldsymbol{p}^T \tag{A299}$$

$$\lim_{k\to\infty} \frac{1}{k} \sum_{i=0}^{k-1} \boldsymbol{p}_0^T \boldsymbol{P}^i \;=\; \boldsymbol{p}^T\;. \tag{A300}$$

A7.2 Q-value Transformations: Shaping Reward, Baseline, and Normalization

The Bellman equation for the action-value function q^π is

$$q^\pi(s,a) \;=\; \sum_{s',r} p(s',r \mid s,a) \left[r \;+\; \sum_{a'} \pi(a' \mid s')\, q^\pi(s',a') \right] . \tag{A301}$$

The expected return at time $t=0$ is:

$$v_0 \;=\; \sum_{s_0} p(s_0)\, v(s_0) \; . \tag{A302}$$

As introduced for the REINFORCE algorithm, we can subtract a baseline v_0 from the return. We subtract the baseline v_0 from the last reward. Therefore, for the new reward $\bar{R}$ we have $\bar{R}_t = R_t$ for $t \leqslant T$ and $\bar{R}_{T+1} = R_{T+1} - v_0$. Consequently, $\bar{q}(s_t,a_t) = q(s_t,a_t) - v_0$ for $t \leqslant T$.
The TD update rules are:

$$q(s_t,a_t) \;\longleftarrow\; q(s_t,a_t) \;+\; \alpha \left(r_t \;+\; \sum_a \pi(a \mid s_{t+1})\, q(s_{t+1},a) \;-\; q(s_t,a_t) \right) . \tag{A303}$$

The δ-errors are

$$\begin{aligned} & R_{t+1} \;+\; \sum_a \pi(a \mid s_{t+1})\, q(s_{t+1},a) \;-\; q(s_t,a_t) \\ &= R_{t+1} \;+\; \sum_a \pi(a \mid s_{t+1})\, (q(s_{t+1},a) - v_0) \;-\; (q(s_t,a_t) - v_0) \\ &= \bar{R}_{t+1} \;+\; \sum_a \pi(a \mid s_{t+1})\, \bar{q}(s_{t+1},a) \;-\; \bar{q}(s_t,a_t) \end{aligned} \tag{A304}$$

and for the last step

$$\begin{aligned} & R_{T+1} \;-\; q(s_T,a_T) \;=\; (R_{T+1} \;-\; v_0) \;-\; (q(s_T,a_T) \;-\; v_0) \\ &= \bar{R}_{T+1} \;-\; \bar{q}(s_T,a_T) \; . \end{aligned} \tag{A305}$$

If we set

$$\bar{q}(s_t,a_t) \;=\; \big\{ q(s_t,a_t) \;-\; v_0 \; , \quad \text{for } t \leqslant T \; . \tag{A306}$$

$$\bar{R}_t \;=\; \begin{cases} R_t \; , & \text{for } t \leqslant T \\ R_{T+1} \;-\; v_0 \; , & \text{for } t = T+1 \; , \end{cases} \tag{A307}$$

then the δ-errors and the updates remain the same for q and $\bar{q}$. We are equally far away from the optimal solution in both cases.
Removing the offset v_0 at the end by $\bar{R}_{T+1} = R_{T+1} - v_0$, can also be derived via reward shaping. However, the offset has to be added at the beginning: $\bar{R}_1 = R_1 + v_0$. Reward shaping requires for the shaping reward F and a potential function Φ [87, 143]:

$$F(s_t,a_t,s_{t+1}) \;=\; \Phi(s_{t+1}) - \Phi(s_t) \; . \tag{A308}$$

For introducing a reward of c at time $t=k$ and removing it from time $t=m<k$ we set:

$$\Phi(s_t) \;=\; \begin{cases} 0 \; , & \text{for } t \leqslant m \; , \\ -c \; , & \text{for } m+1 \leqslant t \leqslant k \; , \\ 0 \; , & \text{for } t > k \; , \end{cases} \tag{A309}$$

then the shaping reward is

$$F\big(s_t,a_t,s_{t+1}\big) \;=\; \begin{cases} 0 \; , & \text{for } t < m \; , \\ -c \; , & \text{for } t = m \; , \\ 0 \; , & \text{for } m+1 \leqslant t < k \; , \\ c \; , & \text{for } t = k \; , \\ 0 \; , & \text{for } t > k \; . \end{cases} \tag{A310}$$

For $k = T$, $m = 0$, and $c = -v_0$ we obtain above situation but with $\bar{R}_1 = R_1 + v_0$ and $\bar{R}_{T+1} = R_{T+1} - v_0$, that is, v_0 is removed at the end and added at the beginning. All Q-values except $q(s_0, a_0)$ are decreased by v_0. In the general case, all Q-values $q(s_t, a_t)$ with $m + 1 \leqslant t \leqslant k$ are increased by c.

Q-value normalization: We apply reward shaping [87, 143] for normalization of the Q-values. The potential $\Phi(s)$ defines the shaping reward $F(s_t, a_t, s_{t+1}) = \Phi(s_{t+1}) - \Phi(s_t)$. The optimal policies do not change and the Q-values become

$$q^{\text{new}}(s_t, a_t) \;=\; q(s_t, a_t) \;-\; \Phi(s_t) \;. \tag{A311}$$

We change the Q-values for all $1 \leqslant t \leqslant T$, but not for $t = 0$ and $t = T + 1$. The first and the last Q-values are not normalized. All the shaped reward is added/subtracted to/from the initial and the last reward.

- The maximal Q-values are zero and the non-optimal Q-values are negative for all $1 \leqslant t \leqslant T$:

$$\Phi(s_t) \;=\; \max_a q(s_t, a) \;. \tag{A312}$$

- The minimal Q-values are zero and all others Q-values are positive for all $1 \leqslant t \leqslant T - 1$:

$$\Phi(s_t) \;=\; \min_a q(s_t, a) \;. \tag{A313}$$

A7.3 Alternative Definition of State Enrichment

Next, we define state-enriched processes $\tilde{\mathcal{P}}$ compared to $\mathcal{P}$. The state $\tilde{s}$ of $\tilde{\mathcal{P}}$ is enriched with a deterministic information compared to a state s of $\mathcal{P}$. The enriched information in $\tilde{s}$ can be computed from the state-action pair $(\tilde{s}, a)$ and the reward r. Enrichments may be the accumulated reward, count of the time step, a count how often a certain action has been performed, a count how often a certain state has been visited, etc. Givan et al. have already shown that state-enriched Markov decision processes (MDPs) preserve the optimal action-value and action sequence properties as well as the optimal policies of the model [34]. Theorem 7 and Corollary 9.1 in Givan et al. proved these properties [34] by bisimulations (stochastically bisimilar MDPs). A homomorphism between MDPs maps a MDP to another one with corresponding reward and transitions probabilities. Ravindran and Barto have shown that solving the original MDP can be done by solving a homomorphic image [99]. Therefore, Ravindran and Barto have also shown that state-enriched MDPs preserve the optimal action-value and action sequence properties. Li et al. give an overview over state abstraction or state aggregation for MDPs, which covers state-enriched MDPs [73].

Definition A14. *A decision process $\tilde{\mathcal{P}}$ is* state-enriched *compared to a decision process $\mathcal{P}$ if following conditions hold. If $\tilde{s}$ is the state of $\tilde{\mathcal{P}}$, then there exists a function $f : \tilde{s} \to s$ with $f(\tilde{s}) = s$, where s is the state of $\mathcal{P}$. There exists a function $g : \tilde{s} \to \mathcal{R}$, where $g(\tilde{s})$ gives the additional information of state $\tilde{s}$ compared to $f(\tilde{s})$. There exists a function ν with $\nu(f(\tilde{s}), g(\tilde{s})) = \tilde{s}$, that is, the state $\tilde{s}$ can be constructed from the original state and the additional information. There exists a function H with $h(\tilde{s}') = H(r, \tilde{s}, a)$, where $\tilde{s}'$ is the next state and r the reward. H ensures that $h(\tilde{s}')$ of the next state $\tilde{s}'$ can be computed from reward r, actual state $\tilde{s}$, and the actual action a. Consequently, $\tilde{s}'$ can be computed from $(r, \tilde{s}, a)$. For all $\tilde{s}$ and $\tilde{s}'$ following holds:*

$$\tilde{p}(\tilde{s}', r \mid \tilde{s}, a) \;=\; p(f(\tilde{s}'), r \mid f(\tilde{s}), a) \;, \tag{A314}$$

$$\tilde{p}_0(\tilde{s}_0) \;=\; p_0(f(\tilde{s}_0)) \;, \tag{A315}$$

where $\tilde{p}_0$ and p_0 are the probabilities of the initial states of $\tilde{\mathcal{P}}$ and $\mathcal{P}$, respectively.

If the reward is deterministic, then $\tilde{p}(\tilde{s}', r \mid \tilde{s}, a) = \tilde{p}(\tilde{s}' \mid \tilde{s}, a)$ and $\tilde{p}_0(\tilde{s}_0, r) = \tilde{p}_0(\tilde{s}_0)$.
We proof the following theorem, even if it has been proved several times as mention above.

Theorem A13. *If decision process $\tilde{\mathcal{P}}$ is state-enriched compared to $\mathcal{P}$, then for each optimal policy $\tilde{\pi}^*$ of $\tilde{\mathcal{P}}$ there exists an equivalent optimal policy π^* of $\mathcal{P}$, and vice versa, with $\tilde{\pi}^*(\tilde{s}) = \pi^*(f(\tilde{s}))$. The optimal return is the same for $\tilde{\mathcal{P}}$ and $\mathcal{P}$.*

Proof. We proof by induction that $\tilde{q}^{\tilde{\pi}}(\tilde{s}, a) = q^{\pi}(f(\tilde{s}), a)$ if $\tilde{\pi}(\tilde{s}) = \pi(f(\tilde{s}))$.
Basis: The end of the sequence. For $t \geqslant T$ we have $\tilde{q}^{\tilde{\pi}}(\tilde{s}, a) = q^{\pi}(f(\tilde{s}), a) = 0$, since no policy receives reward for $t \geqslant T$.